10/23/17 Chlorophyll Samples from Falls Lake (collected 10/19 & 10/20)

Sample ID	Weight (g) foiled flask	Weight (g) flask	Weight (g) sample
04	~~630.56~~ 660.52	473.81	186.75
07	~~756~~.43 757.19	473.~~86~~74	283.45
05	783.03	473.88	309.15
03	692.78	473.79	218.99
12	837.82	473.84	363.98
14	702.89	473.68	229.21
15	739.62	473.84	265.78
01	761.88	474.31	287.57
50	566.03	473.92	92.11
92	575.73	473.71	102.02
91	599.38	473.74	125.64
~~10~~ 44	561.02	473.93	87.09
Blank 01	727.18	473.99	253.17

To Page No. ____

ssed & Understood by me,

Date

Invented by:

Recorded by:

Date

	CHAIN OF CUSTODY FORM (Original Stays With Samples)			US EPA Experimental Stream Facility 1003 US Highway 50, Milford, Ohio 45150 Phone # 513-965-4805 FAX # 513-965-4806			Originator:		
							Ship Date: 10/24/17		Contact:
							Destination:		Phone #:
							Recipient:		E-Mail:
							Phone #:		

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	CrossID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Analyses Required
2017/018						DI	Field BLK	TN02, TN02-3, TNH4, TRP, TP, TN
2017/019	12:38:00	FL	05	0.1	1	SW	UNK	"
2017/019	12:54:00	"	12	0.1	1	SW	UNK	"
2017/019	13:42:00	"	01	0.1	1	SW	UNK	"
2017/019	14:20:00	"	07	0.1	1	SW	UNK	"
2017/019	14:52:00	"	15	0.1	1	SW	UNK	"
2017/019	15:13:00	"	14	0.1	1	SW	UNK	"
2017/019	15:35:00	"	03	0.1	1	SW	UNK	"
2017/019	16:04:00	"	04	0.1	1	SW	UNK	"
2017/020	09:30:00	"	991	0.1	1	SW	UNK	"
2017/020	09:50:00	"	992	0.1	1	SW	UNK	"
2017/020	09:09:00	"	50	0.1	1	SW	UNK	"
2017/020	10:30:00	"	*40	0.1	1	SW	UNK	"
2017/024	12:05:00	"			1	House DI	UNK	"
2017/018		"			1	DI	Field Blank	TOC
2017/019	12:38:00	FL	05	0.1	1	SW	UNK	TOC
2017/019	12:54:00	FL	12	0.1	1	SW	UNK	TOC
2017/019	13:42:00	FL	01	0.1	1	SW	UNK	TOC
2017/019	14:20:00	FL	07	0.1	1	SW	UNK	TOC
2017/019	14:52:00	FL	15	0.1	1	SW	UNK	TOC
2017/019	15:13:00	FL	14	0.1	1	SW	UNK	TOC
2017/019	15:35:00	FL	03	0.1	1	SW	UNK	TOC
2017/019	16:04:00	FL	04	0.1	1	SW	UNK	TOC
2017/020	09:30:00	FL	991	0.1	1	SW	UNK	TOC
2017/020	09:50:00	FL	992	0.1	1	SW	UNK	TOC
2017/020	09:09:00	FL	50	0.1	1	SW	UNK	TOC
2017/020	10:30:00	FL	*40	0.1	1	SW	UNK	TOC
2017/024	12:05:00		Date:	Time:		House DI	Date:	TOC Time:

* This is really sample FL.44 (side), transcribing error. Samantha was notified of error.

CHAIN OF CUSTODY FORM
(Original Stays With Samples)

US EPA Experimental Stream Facility
1003 US Highway 50, Milford, Ohio 45150
Phone # 513-965-4805
FAX # 513-965-4806

Originator:		
Ship Date:	10/24/17	Contact:
Destination:		Phone #:
Recipient:		E-Mail:
Phone #:		

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	Cross ID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Analyses Required	*Volume of liquid sample filtered (mL)	
20171019	12:38:00	FL	05	0.1	1	SW	UNK	Chlorophyll	309.15	
20171019	12:54:00	FL	12	0.1	1	SW	UNK	"	353.98	
2017 1019	13:42:00	FL	01	0.1	1	SW	UNK	"	287.57	
20171019	14:20:00	FL	07	0.1	1	SW	UNK	"	283.45	
2017 1019	14:52:00	FL	15	0.1	1	SW	UNK	"	265.78	
2017 1019	15:13:00	FL	14	0.1	1	SW	UNK	"	229.21	
2017 1019	15:35:00	FL	03	0.1	1	SW	UNK	"	218.99	
2017 1019	16:04:00	FL	04	0.1	1	SW	UNK	"	186.71	
20171020	09:30:00	FL	991	0.1	1	SW	UNK	"	125.64	
20171020	09:50:00	FL	992	0.1	1	SW	UNK	"	102.02	
20171020	09:09:00	FL	50	0.1	1	SW	UNK	"	92.11	
20171020	10:30:00	FL	40	0.1	1	SW	UNK	"	87.09	
20171018		FL					Field BLK	"	253.19	

Relinquished By:	Date:	Time:	Received By:		

GAS TRAP DATA SHEET

Lake: FALLS LAKE

Site	Lat	Long	Deployment time/date	Retrieval time/date	Volume recovered (mL)	Exetainer codes	Depth
24	35.941978	78.588026	10/18/17 10:55	11:19 10/19/17	0		6.5
16	35.95672	78.59041	11:11	11:47	14		20
2	35.97716	78.58900	11:20	12:02	12		10
5	35.97015	78.62127	11:40	12:38	0		9
12	35.965409	78.63397	11:50	17:54	10		14
13	36.00364	78.64627	12:05	13:19	75 → 25 lost	BT.1 : BT.2	17
1	36.00205	78.63436	12:15	13:42	30		5
7	36.00547	78.66783	12:30	14:20	2		25
11	36.01934	78.71705	13:05	14:41	1		9
15	36.02274	78.71465	13:15	14:52	8		14
14	36.04627	78.72021	13:30	15:13	0	Trap was on bottom	2
3	36.03246	78.73684	13:40	15:35	6 : 20 : 50 lost		2.5
8	36.01896	78.74879	13:50	15:53	25 ; 17	Funnel Disconnected BT.1 BT.2	10.5
4	36.01256	78.76085	14:00	16:04	25 ; 8	BT.1	2.5
10	36.03985	78.75606	14:15	16:28	0		2.5
45	36.04410	78.75816	14:20	16:41	6		5.5
57	36.04850	78.75238	14:35	16:56	12		5.0
49	36.05128	78.76682	14:40	17:11	20	BT.1 Dumped	7.0
~~55~~	36.05322	78.76163	14:50	17:23	14		5.0
54	36.05785	78.76522	14:55	10/20 12:25	0		3.5
43	36.06151	78.77318	15:05	10/20 12:00	25, 17		3.5
991	—	36.07039	15:50	10/20 09:30	25, 25, 25, 140, 140, 140, 140, 140, 115		3.5
992	—	36.07016	15:55	10/20 09:50	25, 2		3.0
56	36.06863	78.78669	16:10	10/20 09:09	25 25 3		2.5
44	36.07218	78.77948	16:30	10/20 ~~10:30~~	6	TRAP on bottom	2.5
58	36.06860	78.77593	16:55	10/20 10:50	25, 12		3.0

21 — —

SONDE DATA SHEET

Lake: Falls Lake Sample Date: 10-19-17 / 10-2-17

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond (µs cm^{-1})	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
24	6.5	0.1	23.1	6.06	70.9	0.089	8.35	6.7	3.8	
16	20	0.1	22.6	~~6.05~~ 6.04	62.8	0.089	8.17	14.0	3.1	
2	10	0.1	22.0	10.47	119.4	0.083	8.17	~~7.89~~ 9.0	4.8	
5	9	0.1	22.16	5.79	65.6	0.088	8.05	7.3	3.1	Blue under water
12	14	0.1	23.63	7.3	85.3	0.087	7.97	3.9	2.9	[redacted]
13	17	0.1	23.19	7.31	85.0	0.085	7.95	10.7	4.6	
~~81~~	5	0.1	23.64	9.54	111.5	0.083	7.98	7.6	3.6	
~~7~~ 21	2.5 4	0.1	22.22	10.28	118.2	0.083	8.01	10.9	4.8	
7	25	0.1	23.07	9.82	114.5	0.085	7.96	7.5	4.0	
11	9	0.1	21.8	11.58	133.5	0.~~09~~1	8.22	7.4	5.0	
5	14	0.1	22.46	11.59	133.9	0.096	8.36	8.1	5.0	
14	2	0.1	21.84	11.52	131.5	0.093	8.43	5.0	6.7	
3	2.5	0.1	23.1	11.79	137.6	0.104	8.45	8.3	10.6	
8	10.5	0.1	21.58	11.72	134.5	0.111	8.46	9.4	6.5	
9% sat 140.0 → 4	2.5	0.1	21.05	~~11.89~~ 12.4	~~134.5~~ 140.0	0.111	8.55	9.2	10.7	
10	2.5	0.1	23.11	11.8	138.0	0.136	8.40	9.0	7.8	
45	5.5	0.1	21.23	12.~~15~~2	138.0	0.137	8.63	7.1	8.7	
57	5.0	0.1	20.9	11.78	132.0	0.148	8.63	7.9	9.7	
49	7.0	0.1	20.9	12.53	140.7	0.180	8.65	11.0	14.0	
55	5.0	0.1	22.36	12.76	147.1	0.194	8.77	13.5	18.4	
10/20/17 50	2.5	0.1	16.25	11.75	119.1	0.295	8.28	23.4	45.0	
991	3.0	0.1	16.42	12.54	128.4	0.353	8.44	27.6	45.5	
592	3.0	0.1	16.6	13.2	139.0	0.332	8.74	10.0	36.6	
44	2.5	0.1	16.6	~~10.63~~	108.0	0.233	8.61	19.2	59.0	
58	3.0	0.1	17.5	11.2	117.2	0.243	8.60	38.2	32.3	
47	5.0	0.1	18.2	11.64	123.6	0.20	8.62	9.4	18.6	
56	5.8	0.1	18.4	11.25	119.8	0.198	8.62	8.7	13.9	
52	2.8	0.1	18.2	11.21	119.2	0.194	8.46	~~8.45~~ 12.6	16.9	
43	3.5	0.1	18.8	11.30	123.0	0.214	8.57	12.3	24.7	
54	3.5	0.1	18.6	11.8	126.5	0.194	8.65	11.7	18.8	

DISSOLVED GAS AND AIR DATA SHEET

Lake:	Falls Lake				Date:	10-19-17		
¥Site	Dissolved gas Exetainer codes		Water Vol/ Helium Vol	Water Temp (degC)	Atmospheric Air Exetainer Codes			BP (mm Hg)
2	DG.1	DG.2 115/25		23.6	AA.1 AA.2	AA.3 AA.4	AA.5	762.8
5	DG.1	DG.2		25.6	—	—	—	
1	DG.1	DG.2		25.8				
7	DG.1	DG.2		24.3				
15	DG1	DG2		23.8				
14	DG.1	DG.2		25.5				
3	DG.1	DG2		26.3				
4	DG.1	DG.2		23.6				
791	DG.1	DG.2		18.5				
50	DG1	DG2		16.6				
992	DG1	DG2		19.6	AA.1 AA.2	AA.3 AA.4	AA.5	762.6
44	DG1	DG2		18.8				

¥Site: all dissolved gas samples collected at 0.1m depth
Boat motor should be off for 2 minutes before sample collection
Goal is to inject 20mL of gas into exetainer

Signature Date

GREENHOUSE GAS ANALYZER AND GPS DATA SHEET

FILE NAMES		
Lake:	Falls Lake	
Sample Date:	10-19-17	10-20-17
GGA file	GGA_2017-10-19_F0000.txt	GGA-2017-10-20-f0000.txt
gps file name	BadElf	

DEPLOYMENT DATA

Sample site	Chamber start time	Bubbling observed (Y/N)	GPS logger started	Chamber volume (L)	CH_4 and CO_2 profile checked
24	~~24:56~~ 11:56:24 (33:05)	N	Y	30	Y
16	11:48:16	~~N~~ Y	Y	30	Y
2	12:05:21	N	Y	30	Y
5	12:41:13	N	Y	30	Y
12	12:56:34	Y	Y	30	Y
13	13:22:07	Y	~~N~~ Y	40	Y
1	13:45:45	N	Y	40	Y
~~21~~	14:02:41	N	Y	40	Y
7	14:22:46	N	Y	40	Y
11	14:44:35	N	Y	50	Y
15	14:54:37	Y	Y	30	Y
14	15:17:32	~~N~~ Y	Y	40	Y
3	15:39:18	Y	Y	40	Y
8	15:55:10	N	Y	40	Y
4	16:11:28	N	Y	40	Y
10	16:30:12	N	Y	35	Y
45	16:41:05	N	Y	40	Y
57	16:59:05	N	Y	40	Y
49	17:12:45	N	Y	50	Y
55	17:25:28	N	Y	40	Y
50	09:12:10	N	Y	40	Y
991	09:29:44	N	Y	40	Y
992	09:54:45	Y	Y	40	Y
44	10:30:48	N	Y	40	Y
58	10:46:50	N	Y	40	Y
47	11:03:29	N	Y	40	Y
56	11:17:15	N	Y	40	Y
52	11:35:56	N	Y	40	Y
43	12:01:00	N	Y	40	Y
54	12:24.00	N	Y	40	Y

Bad Elf time 15:34

10/20/17

Near I 85 Ambient CH_4 elevated

BAD ELF = LGR

10/20/17 Time 16:27:00 = 12:25:40

10/18/17 YSI 6600 Sonde

Pre deployment

	T	Cond	pH	ORP	NTU	Chl	DO
DI	24.84	0.004	8.7	57.3	0.1	0.4	2.63

NTU	DI
	0.1

124 NTU
128.1

pH	Buffer	pH	T
	0	8.7	24.84
	7	7.06	21.69
	10	10.2	21.62

Conductivity	standard	Cond ms/cm
	0	0.004
	1413	1453

Temp	YSI	Thermocouple
	21.76	21.9

SONDE DATA SHEET

Lake: Falls Lake Sample Date: 10-19-17 / 10-20-17

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond (µs cm^{-1})	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
24	6.5	0.1	23.1	6.06	70.9	0.089	8.35	6.7	3.8	
16	20	0.1	22.6	~~6.05~~ 5.4	62.8	0.089	8.17	14.0	3.1	
2	10	0.1	22.0	10.47	119.4	0.083	8.17	~~7.8~~ 9.0	4.8	
5	9	0.1	22.16	5.79	65.6	0.088	8.05	7.3	3.1	Blue under water
12	14	0.1	23.63	7.3	85.3	0.087	7.97	3.9	2.9	~~Blue~~
13	17	0.1	23.19	7.3)	85.0	0.085	7.95	10.7	4.6	
~~81~~	5	0.1	23.64	9.54	111.5	0.083	7.98	7.6	3.6	
~~7~~ 21	~~2~~ 5 4	0.1	22.22	10.28	118.2	0.083	8.01	10.9	4.8	
7	25	0.1	23.07	9.82	114.5	0.085	7.96	7.5	4.0	
11	9	0.1	21.8	11.58	133.5	0.~~84~~	8.22	7.4	5.0	
5	14	0.1	22.46	11.59	133.9	0.096	8.36	8.1	5.0	
14	2	0.1	21.84	11.52	131.5	0.093	8.43	5.0	6.7	
3	2.5	0.1	23.1	11.79	137.6	0.104	8.45	8.3	10.6	
8	10.5	0.1	21.58	11.72	134.5	0.111	8.46	9.4	6.5	
4 (9% sat 140.0)	2.5	0.1	21.05	~~11.8~~ 12	~~134.5~~ 140.0	0.111	8.55	9.2	10.7	
10	2.5	0.1	23.11	11.8	138.0	0.136	8.40	9.0	7.8	
45	5.5	0.1	21.23	12.~~15~~2	138.0	0.137	8.63	7.1	8.7	
~~57~~	5.0	0.1	~~20.9~~	11.78	132.0	0.~~14~~8	8.63	7.9	9.7	
49	7.0	0.1	20.9	12.53	140.7	0.180	8.65	11.0	14.0	
55	5.0	0.1	22.36	12.76	147.1	0.194	8.77	13.5	15.4	
10/20/17 50	2.5	0.1	16.25	11.75	119.1	0.295	8.28	23.4	45.0	
991	3.0	0.1	16.42	12.54	128.4	0.353	8.44	27.6	45.5	
592	3.0	0.1	16.6	13.2	139.0	0.332	8.74	10.0	36.6	
44	2.5	0.1	16.6	~~10.6~~	108.0	0.233	8.61	19.2	59.0	
58	3.0	0.1	17.5	11.2	117.2	0.243	8.60	38.2	32.3	
47	5.0	0.1	18.2	11.64	123.6	0.20	8.62	9.4	18.6	
56	5.5	0.1	18.4	11.25	119.8	0.198	8.62	8.7	13.9	
52	2.8	0.1	18.2	11.21	119.2	0.194	8.46	~~8.0~~ 12.6	16.9	
43	3.5	0.1	18.8	11.30	123.0	0.214	8.57	12.3	24.7	
54	3.5	0.1	18.6	11.8	126.5	0.194	8.65	11.7	18.8	

Chamber LGR

Date 10/13/17
Cal cylinder serial # CC163816
Dilution system serial # TEI EPA # A20808
LGR file 994_2017-10-13_f0000.txt

Zero Air Flow (Lpm)	Cal Gas Flow Set (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.02	0.~~02~~ 78	13:12	
9	7 ~~9.4~~	2.1	20.9	2.3	23.3	12:41	
5.3	7 ~~9.4~~	3.6	35.5	3.9	39.1	12:58	
5.3	10 ~~13.4~~	5.1	50.7	5.4	53.9	12:44	
3.25	15 ~~20.2~~	12.4	124.1	12.9	128.5	12:47	
3.25	20 ~~26.9~~	16.5	165.5	16.8	169.2	13:03	
3.25	30 ~~40.3~~	24.8	248.3	24.7	251.4	12:54	
3.25	60 ~~80.7~~	49.6	496.5	46.9	481.8	12:35	
3.25	70 ~~94.1~~	57.9	579.3	53.6	563.2	13:07	
3.25	90 ~~121.0~~	74.4	744.8	4.75	716.7	13:09	

Signature